

CSIRO Publishing

Australian *Journal* of Soil Research



VOLUME 40, 2002

© CSIRO 2002

An international journal for the publication of
original research into all aspects of soil science

All enquiries and manuscripts should be directed to:

Australian Journal of Soil Research
CSIRO Publishing
PO Box 1139 (150 Oxford St)
Collingwood, Vic. 3066, Australia



Telephone: +61 3 9662 7628
Fax: +61 3 9662 7611
Email: publishing.sr@csiro.au

Published by CSIRO Publishing
for CSIRO and the Australian Academy of Science

www.publish.csiro.au/journals/ajsr

Interpretation of morphological features in a salt-affected duplex soil toposequence with an altered soil water regime in western Victoria

J. Brouwer^{AC} and R. W. Fitzpatrick^B

^ABrouwer Envir. & Agric. Consultancy, Wildekamp 32, 6721 JD Bennekom, The Netherlands; formerly at the Centre for Land Protection Research, Department of Conservation and Natural Resources, Osborne Street, Bendigo, Victoria 3550, Australia, and the Department of Soil Science and Geology, Wageningen University and Research Centre, The Netherlands; Visiting Scientist at CSIRO, Glen Osmond, SA, Australia.

^BCSIRO Land and Water, and CRC for Landscape Environments and Mineral Exploration, Private Bag No. 2, Glen Osmond, SA 5064, Australia.

^CCorresponding author; email: brouwbar@bos.nl

Abstract

This paper is the first of two describing how soil macromorphological and chemical data can be combined with a minimum of hydrological data to distinguish between, and to quantify, past and present hydrological processes. These processes are relevant to both waterlogging and dryland salinity. The purpose of this first paper is to establish a methodological framework. It also describes the initial interpretation of the macromorphological features of the toposequence studied at Gatum on the Dundas Tablelands in western Victoria. A modified version of the soil feature–system–domain grouping method was used. Macromorphological data combined with only limited piezometric data showed that:

- (1) The soil feature–system–domain grouping method makes it possible to distinguish between the effects of past and present hydrological processes on soil macromorphology at Gatum.
- (2) Waterlogging of the surface horizons at Gatum is often caused by perching of soil water within the B-horizon (as opposed to on top of the B-horizon). Changes in soil structure and in colour of cutans and mottles can be an indicator of this first restricting layer.
- (3) It is likely that interpedal cracks and old tree root holes act as preferred paths for water to flow through this first restricting layer.
- (4) A second fresh perched water table can occur on top of the pallid zone. Where the pallid zone reaches close to the surface the two perched water tables may merge and cause a local increase in waterlogging, as indicated by local soil morphology. When this occurs, hillside seeps can occur quite high up on the slopes, even when there is no apparent irregularity in surface topography.
- (5) The permanent saline water table occurs on top of the bedrock and causes salting problems where it comes too close to the soil surface. Salting problems at the bottom of a slope are more severe where fresh perched water tables increase waterlogging.

On the basis of these findings the suitability of various management options to reduce waterlogging and salinisation is discussed. Further findings regarding restricting layers, flow paths through the soils, and relations between duration of saturation and soil morphological features, are discussed in a companion paper (by J. Brouwer and R. W. Fitzpatrick, pp. 927–946 in this issue).

Additional keywords: soil colour, soil-water-landscape models, duplex soils, Dermosols, Chromosols, Hydrosols.

Introduction

Waterlogging or saturation of topsoils is a major constraint to agricultural production in southern Australia. So is dryland salinity. To understand the occurrence and driving mechanisms of seasonal waterlogging and salinity, knowledge of soil hydrology and its relationship to soil macromorphological features is essential. For example, soil colour features can provide a straightforward means to recognise or predict salt-affected, waterlogged wetlands caused by poor drainage (Fitzpatrick *et al.* 1996, 1999; Brouwer and

Fitzpatrick 1998), providing an alternative to the difficult and expensive process of documenting water table depths to estimate water duration in soils (Cox *et al.* 1996). Visual indicators may be obvious (e.g. white salt accumulations on soil surfaces) or subtle (e.g. subsoil mottling patterns). Macromorphological data can be collected during a single visit and can be considered a reliable measure of average hydrological conditions. These characteristics can be a reflection of past as well as present hydrology, particularly in the ancient landscapes common in Australia. Soil mineralogical and soil chemical data, however, can be used to distinguish between the influences of present and past hydrological processes on soil macromorphology (Boulet *et al.* 1982; Fritsch *et al.* 1992; this paper). The necessary samples can be collected during a single visit and results can be extended to other catchments.

In this paper we present the results of the study of a toposequence of Typic Plinthoxeralfs–Aquic Natriferalfs–Typic Plinthoxeralfs–Typic Natraqualfs in western Victoria. The original broad objective of the study was to find an economically attractive means of reducing groundwater recharge under pastures, primarily through increasing pasture production and thus evapotranspiration. However, initial examination of these duplex soils found that their hydrological behaviour did not conform to the accepted ‘norm’ for such texture-contrast soils in Australia, i.e. the occurrence of ponding on top of the B-horizon. Consequently, these findings led to a more detailed study of the local relationships between geology, soil profile macromorphology, soil chemistry, and hydrology. During this study, comparisons were also made of the advantages and disadvantages of each of the associated disciplines in improving understanding of soil-related processes. Results of macromorphological interpretations and a summary of hydrological findings are presented. A second paper deals with detailed piezometric observations and with quantitative relationships between soil colour and duration of contemporary waterlogging (Brouwer and Fitzpatrick 2002, this issue).

Materials and methods

Site description

Soil and hydrological data were collected at Gatum, on the eastern Dundas Tableland in western Victoria, which lies approximately 300 km west of Melbourne (Fig. 1) at 200–300 m above sea level. The Tableland is a flat, dissected surface with deeply weathered regolith. Valley depths are of the order of 20 m (Fig. 1), with maximum slopes between 6 and 8°. Ongoing weathering, possibly since the Permian, has produced deep weathering profiles, with ferricretes and ferruginous gravelly soils overlying mottled and pallid zones that extend to tens of metres depth. Since emplacement, the area has been tectonically stable except for updoming in the order of 240 m (Joyce 1991). Volcanic centres have not been widely recognised, with the exception of the Glendinning Ignimbrite, which contains abundant large locally derived clasts near the base (Simpson 1998).

At Gatum itself the underlying geology is no longer called rhyolite (Spencer-Jones 1965; Sibley 1967) but the Gatum ignimbrites of the Rocklands Volcanic Group that erupted onto the land surface around 410 Ma (Early Devonian) (Cayley and Taylor 1997; Simpson 1997, 1998; Quinn 1997). Depth to fresh basement rock at Gatum varies from more than 20 m to approximately 12 m (Lewis 1985). Above the fresh ignimbrites there is a zone of weathered rock, which, although of low hydraulic conductivity, acts as the main aquifer (Fig. 2). Because of this low hydraulic conductivity, groundwater hydraulic heads occur on top of the ignimbrites, such that groundwater catchment boundaries generally coincide with surface catchment boundaries; the groundwater regime is of the local discharge type (Lewis 1985). This means that local changes in recharge should have an effect on local groundwater levels and salinity, and not be swamped by inflow of groundwater from outside the catchment (Jenkin 1981). Recent publications by Nathan (1999) and Dahlhaus *et al.* (2000), putting the salinity problems on the Tableland in a wider historical perspective, are discussed in Brouwer and Fitzpatrick (2002, this issue).

Deep weathering of the ignimbrites has resulted in the accumulation of layer silicates, which grade upwards to form a pallid zone (Fig. 2). In the upper part of the pallid zone, the weathering has led to white-

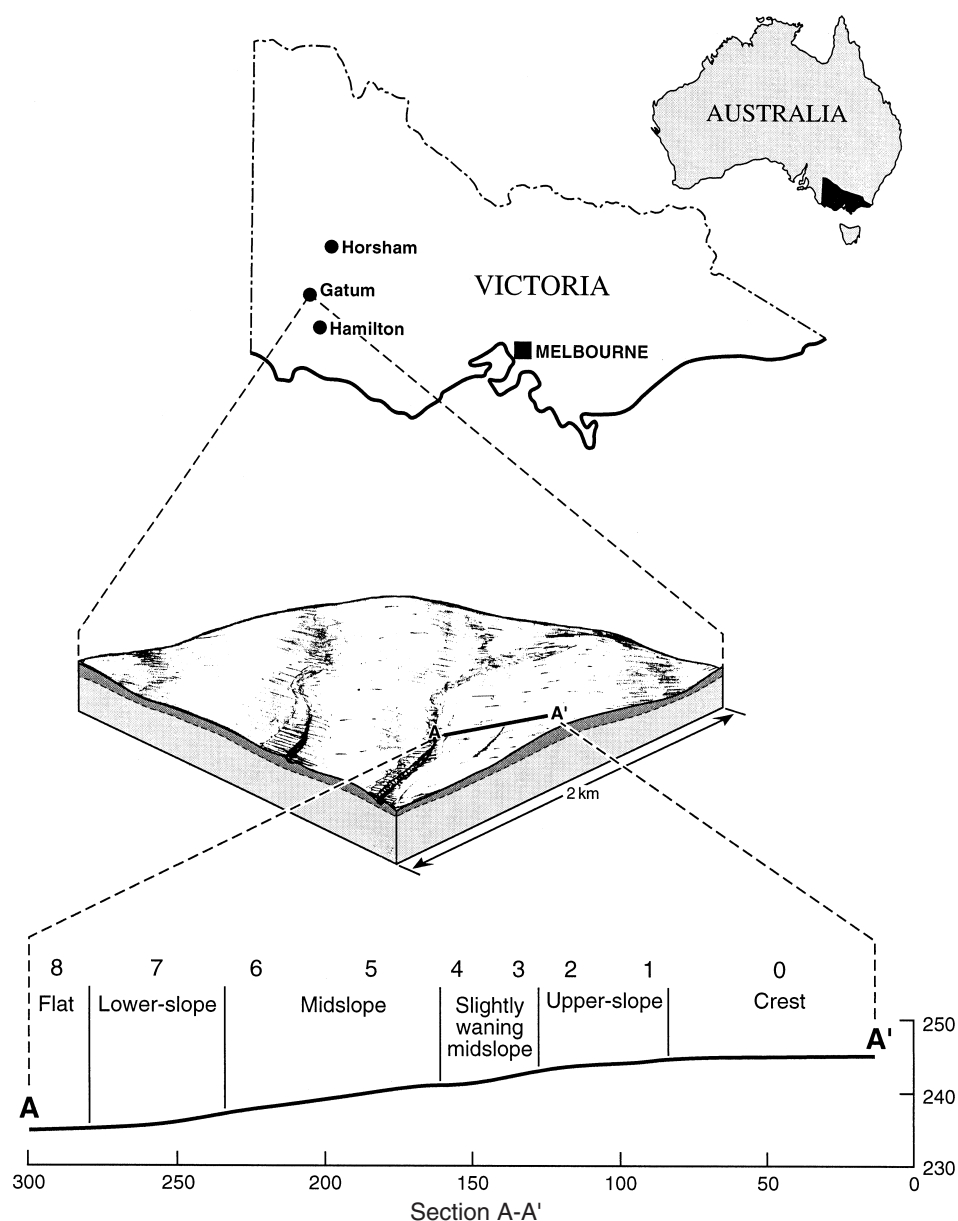


Fig. 1. Location of the Gatum study site and block diagram on the Dundas Tableland, western Victoria, Australia. Numbers indicate piezometer installations and soil profile pit positions.

coloured clay with some sporadic remnant quartz crystals from the original parent material. It is in the pallid zone that cyclic salts have accumulated over thousands of years, now forming a salt store of the order of 500 t/ha (Brouwer and Van de Graaff 1988). Above the pallid zone there may be a zone of yellow or reddish yellow, coarse sandy clay loam, with weakly platy structure and of uncertain origin. Either above this or directly on top of the pallid zone, there is a mottled plinthite zone with strongly platy structure. Of the 'ferricrete hardcaps', formerly present above the mottled plinthite zone, fragment sizes present are mostly gravel, and sometimes stone size (stone dimensions can exceed 0.6 m).

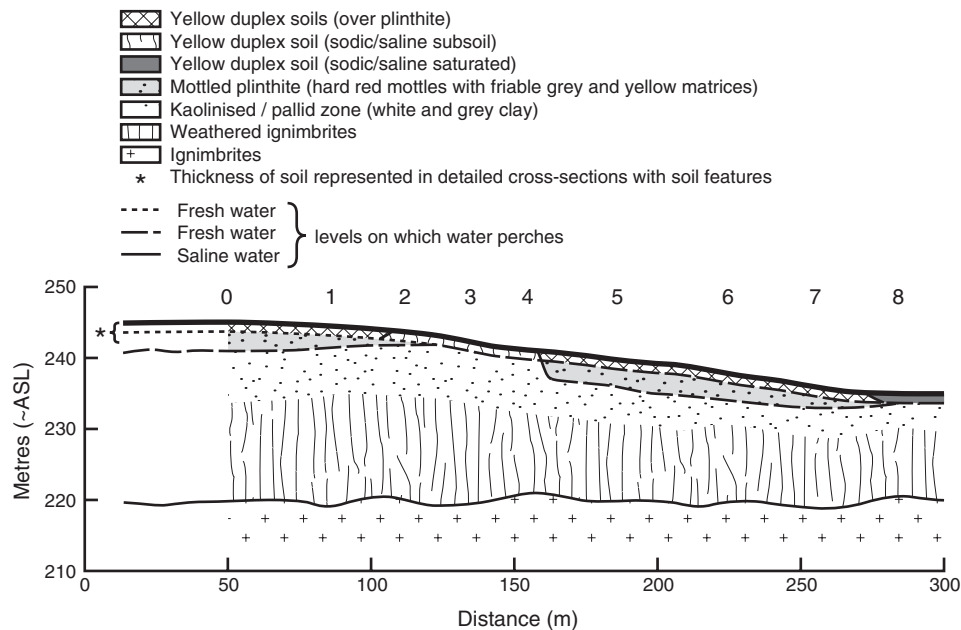


Fig. 2. Geological-pedological cross-section at Gatum.

Like similar landscapes with remnants of lateritic paleosols in nearby South Australia (Milnes *et al.* 1985), the Dundas Tableland has probably undergone a number of weathering, erosion, and deposition cycles, and possibly faulting during tectonic uplift. This has resulted in all of the mentioned paleohorizons outcropping in some part of the landscape, and acting as parent material for soil development. The most recent, and still ongoing, period of soil development has resulted in mainly yellow duplex or texture-contrast soils (Gibbons and Downes 1964). As a consequence of its complicated genesis, the present landscape is considered to be pedologically complex; Fig. 2, a geological-pedological cross-section, illustrates this complexity.

The present climate at Gatum is Mediterranean. Annual rainfall averages approximately 630 mm, mostly occurring in May to October, when there are also occasional frosts. Average annual pan evaporation is approximately 1400 mm. The vegetative cover before clearing for agriculture and development was a *Eucalyptus camaldulensis* woodland. This was at first replaced using mostly annual pasture species, with perennial species being used over the last 15–20 years.

Toposequence description and monitoring

A toposequence at Gatum was described, monitored, and analysed using a modified version of the structural approach method described by Fritsch and Fitzpatrick (1994), which in turn is based on Boulet *et al.* (1982) and Fritsch *et al.* (1992). Details of the method are given by Brouwer and Fitzpatrick (2000), who also included various modifications of the original method, such as the use of spreadsheet software to facilitate data handling.

In summary, the structural approach commences with the selection of representative locations along a toposequence. Toposequence surface topography is recorded using a surveyor's level. At a representative location in each part of the toposequence a profile pit is dug, the profile is described using standard methods (e.g. McDonald *et al.* 1990; Soil Survey Staff 1993) and preferably photographed, and soil samples are taken for laboratory analysis. In addition, piezometers are installed and monitored. In the office, cross-sections of the toposequence are drawn featuring the values of individual, or sometimes several, soil properties. Relevant soil properties include texture, coarse fragments, structure, matrix colour, mottling, and chemical and mineralogical properties. Almost always the vertical topographical scale is exaggerated

relative to the horizontal, to ensure the cross-sections are both legible and fit on one page. Quite often the scale of the soil depth is exaggerated even further, to ensure that thin layers or horizons show up (see Fig. 3). Boundaries are drawn grouping locations with similar values for the soil properties concerned, irrespective of what horizon or profile those values may occur in. This is because the structural method is concerned with the 2- or 3-dimensional spatial arrangement of geological and pedological processes and their relative intensity, or with soil (or soil property) layers, and not with 1- or 2-dimensional horizon arrangements. The horizon concept is too restrictive in this context.

Each mapping unit so delineated is called a soil feature. A soil feature thus represents a limited range of one or more soil properties. Note that soil depth exaggeration can cause soil feature boundaries to slope in the wrong direction, i.e. inwards instead of more or less parallel to the topographic slope. For features such as layers that restrict water flow it may therefore be useful to also draw a cross-section without the extra exaggeration of the soil depth scale (compare Fig. 3d in this paper with Fig. 2 in Brouwer and Fitzpatrick 2002, this issue).

Maps of different types of soil features are compared. The key soil features that help recognise and explain soil formation and interactions between different parts of the toposequence are grouped into the same soil systems using concordant relationships: where there is a concordant relationship spatial distributions and boundaries mostly coincide, and hydrological processes, chemical processes, and/or parent material will be the same. Soil features are separated into different soil systems using discordant relationships; in such cases spatial distributions show no or only partial overlap, boundaries do not coincide but touch or cross each other, and processes and/or parent material will be different. Similarly, the soil systems are grouped into domains (see Fritsch *et al.* 1992; Fritsch and Fitzpatrick 1994; Brouwer and Fitzpatrick 2000). It is emphasised that, as indicated for the discordant relationships, soil features belonging to different systems or domains may overlap spatially.

At Gatum, 8 locations along a representative toposequence were selected for piezometer installation along the top of the B-horizon in 1987 (Fig. 2). At a ninth location, on the adjacent broad crest, piezometers had already been installed to a number of depths in 1985, the deepest going to 19 m (see Brouwer and Van de Graaff 1988). Details of piezometer installation can be found in Brouwer and Fitzpatrick (2000, 2002 this issue). Height and duration of saturation were measured in all the piezometers during autumn–winter–spring of 1987 and 1988. Only limited use is made of the piezometric data in this paper.

In May 1989 all profiles along the toposequence were described using Soil Survey Staff (1993). Comprehensive profile descriptions were made for all typical profiles and abbreviated descriptions were made for nearby similar profiles. Profiles were also photographed for use in further interpretation. Field textures were determined according to Northcote (1979). Laboratory analyses were carried out using methods described in Rayment and Higginson (1992). A deep (>3 m) profile pit was excavated on the broad crest in 1985 (Brouwer and Van de Graaff 1988). Its profile description was adapted to conform with descriptions of all the other profiles. Soil features, systems, and domains in the toposequence were identified, and then represented graphically using the method of Rinder *et al.* (1994). A new pattern was developed by Rinder to represent cutans (see Fig. 3).

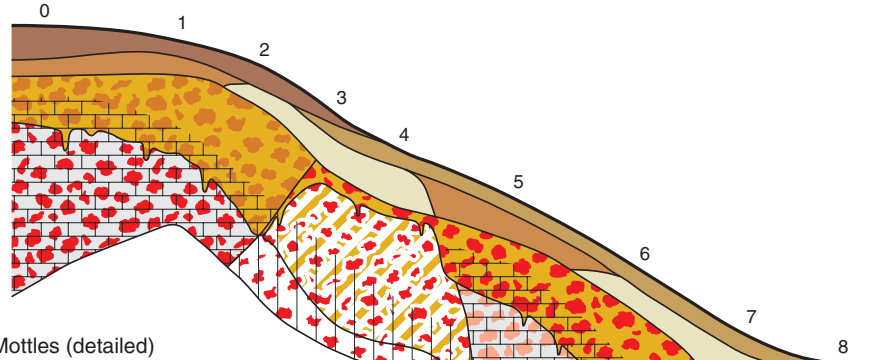
Results

Piezometer data shown in Fig. 4 indicate the presence of 3 water tables: one intermittent water table that is related to a less permeable layer at about 0.8–1.0 m depth, well into the B-horizon; one intermittent water table perched on top of the pallid zone; and a permanent, saline water table on top of the bedrock (Fig. 2 and 3d; see also Brouwer and Van de Graaff 1988 and Fig. 2 in Brouwer and Fitzpatrick 2002, this issue).

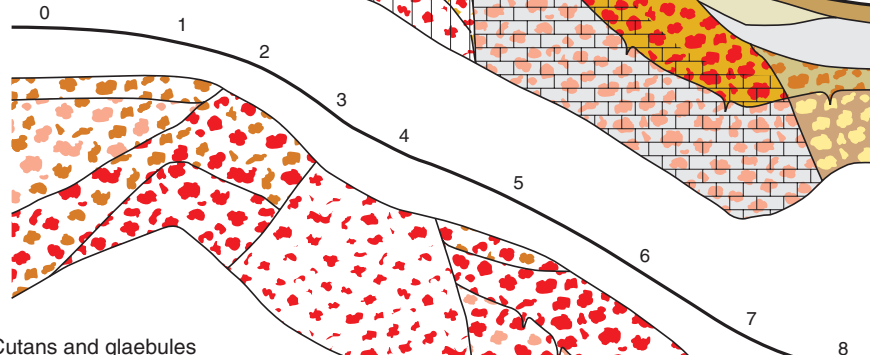
Soil features

Selected morphological and chemical properties of the 4 typical profiles G-0, G-3, G-5, and G-8 are presented in Table 1. Full profile descriptions and colour photographs of the typical profiles, and partial descriptions of related profiles, can be found in Brouwer and Fitzpatrick (2000). The results of the mapping of soil features of the toposequence, as defined by Fritsch *et al.* (1992) and Fritsch and Fitzpatrick (1994), are presented in Fig. 3a–c. Definitions of the various soil features mapped in Fig. 3 are given in Table 2. The following comments are made on the distribution patterns in Fig. 3.

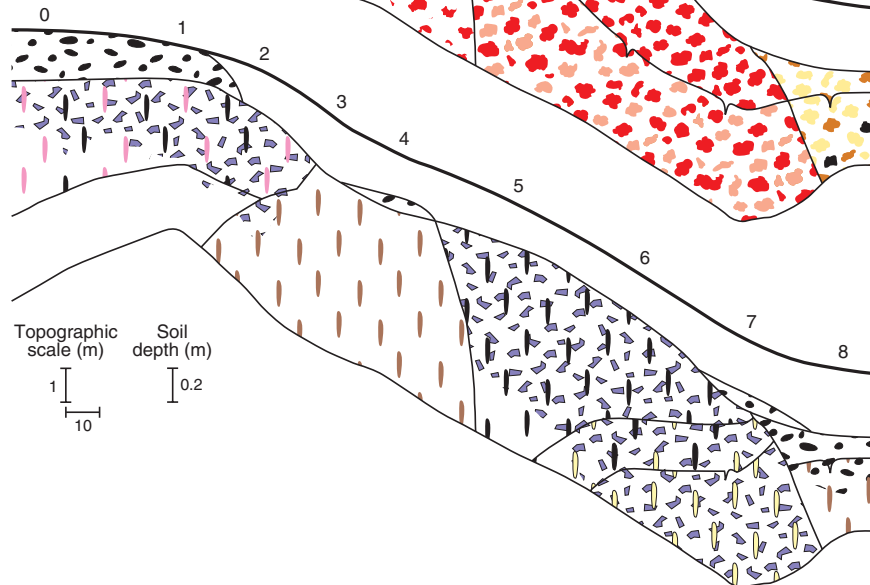
(a) Matrix colours, dominant mottles and structure



(b) Mottles (detailed)



(c) Cutans and glaebules



Topographic
scale (m)
1
10

Soil
depth (m)
0.2

Fig. 3. Soil features at Gatun. Numbers indicate piezometer installations and soil profile pit positions.

A- and E-horizons

The A-horizons are relatively pale in profiles G-3 through to G-8, being dark grey brown rather than dark brown (Fig. 3 and Table 1). This indicates an increase in temporary

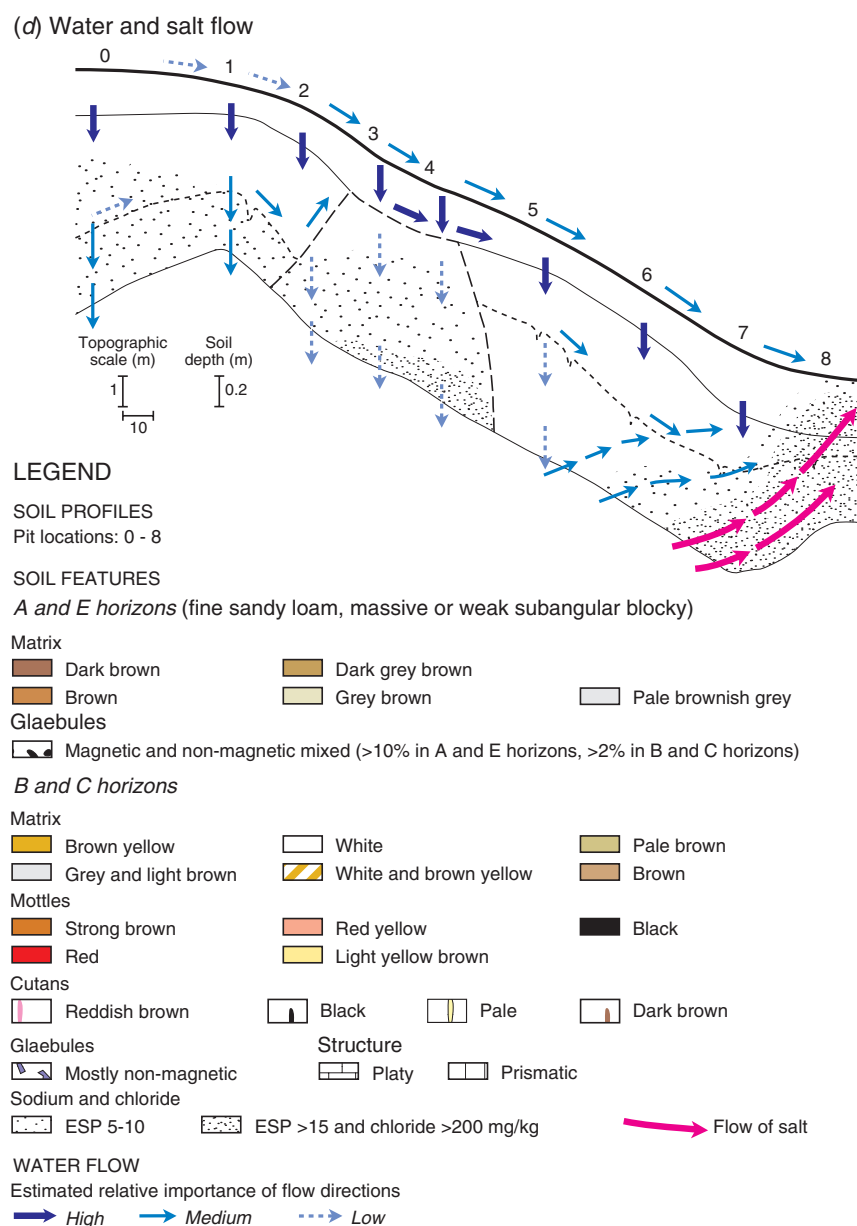


Fig 3. (continued)

saturation of the A-horizon, going from the crest down to lower parts of the landscape. Interflow through the A-horizon will accumulate and persist for longer within the lower parts of the landscape, leading to paler A-horizon colours. However, the colours of the E-horizon show a more complicated picture; a pale colour (and long-lasting saturation) of the E-horizon is found not only on the flat and on the lowest part of the slope, but also in profiles G-2 to G-4, rather high up (grey brown and pale brownish grey moist colours,

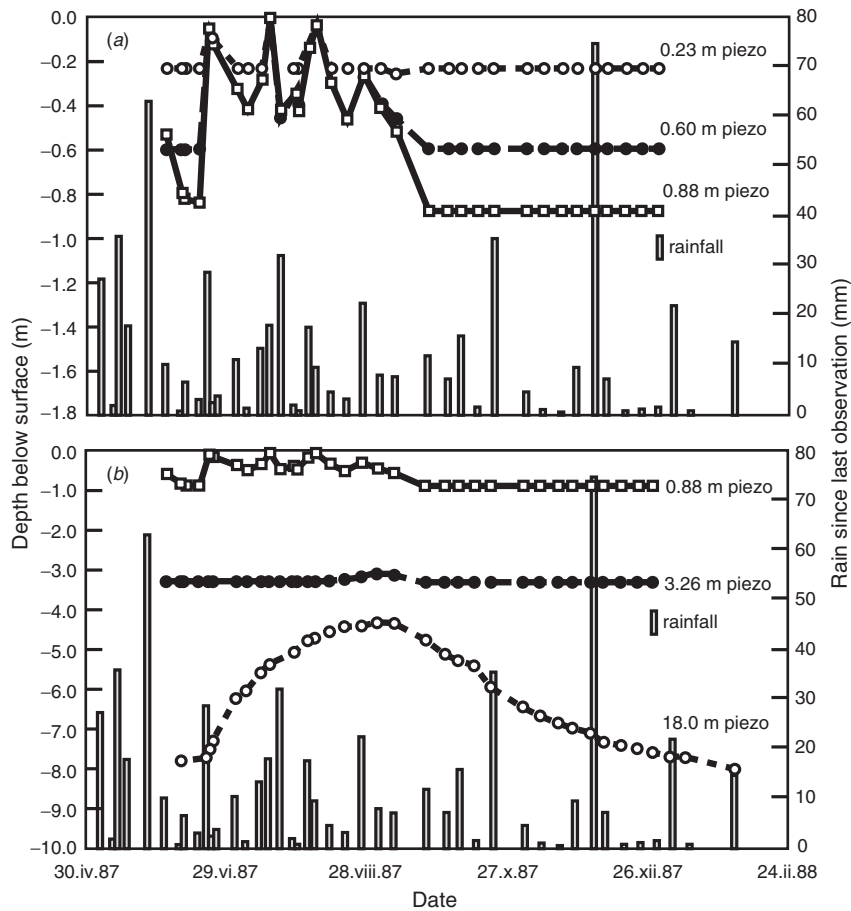


Fig. 4. Piezometer levels on broad crest under conventionally managed phalaris pasture, Gatum, 1987: (a) near the surface, (b) near the surface and at depth. Horizontal line indicates no water in piezometer.

Fig. 3a and Table 2). This is somewhat unexpected, as there is hardly any change in slope of the soil surface in that section of the toposequence. However, as is discussed below, the nature of the underlying B-horizon proved to be different at G-2 to G-4. In addition, the slope of the top of the B-horizon appears to be concave there. This will retard lateral flow over the top of the B-horizon and increase waterlogging and bleaching of the E-horizon.

That the top of the B-horizon is concave at G-3 to G-4 is supported by the presence of ironstone glaeboles or gravel in the E-horizon of profile G-4 (Fig. 3c). The glaeboles appear to have accumulated there, from sources upslope, as a result of slope processes; no ironstone glaeboles are found elsewhere in profile G-4, nor in G-3. This indicates that the slope of the top of the B-horizon is indeed concave in that part of the toposequence. Similarly, ironstone glaeboles have accumulated where the slope of the top of the B-horizon becomes concave in profile G-7. More than 10% ironstone glaeboles are also found in the A- and E-horizons on the crest and upper-slope (G-0 and G-1). Here the ironstone glaeboles are remnants of weathering and erosion of the lateritic layers formerly overlying the local

subsoils. If ironstone glaeboles were currently being formed one would expect to find them in all E-horizons, i.e. also in profiles G-2, 3, 5, 6, and 8.

A- and E-horizon field textures are all rather similar (fine sandy loam) and have not been mapped separately. A- and E-horizon structures have also not been mapped; structure of the topsoil is almost everywhere massive (but porous) to weak subangular blocky. Only at G-8 the A-horizon structure is crumb over a massive E-horizon. Strong brown mottles in the A- and E-horizons showed no clear pattern.

B- and C-horizons

Parent material for the B-horizons along the toposequence can be divided into 4 units on the basis of C-horizon colour (Fig. 3a): grey and light brown with red mottles on the crest and upper slope; white at the slightly waning midslope; grey and light brown with red mottles again under the rest of the midslope and under the lower-slope; and brown on the flat. These units represent respectively a fossil mottled zone ('plinthic on crest'), a fossil pallid zone, another fossil mottled zone ('plinthic on slope'), and alluvial/colluvial deposits on the valley flat. In the 3 parent materials along the slope, a (dark) yellowish brown/brownish yellow B-horizon has formed and may still be forming. Its formation has progressed deeper into the parent material on the shoulder than on the flat: cf. sequence G-0 with G-2. It has also progressed more deeply in the former mottled zones than in the former pallid zone.

Mottle colour varies greatly, both within and between soil profiles (Fig. 3b). The transformation sequence of the mottles in the plinthite, as soil formation progresses downward in the profile, appears to be red (2.5YR, as found at depth), to red with a thin brown rind, to reddish yellow (5YR), to strong brown (7.5YR and even 10YR) near the top of the B-horizon (Fig. 3b, c, and Table 2). Reddish yellow mottles in the lower reaches of profiles G-5 to G-7 were either never fully red (formed under less oxidising circumstances closer to the groundwater table at that time), or they are the first transformations of red mottles caused by (perched) water coming in laterally and from below (see Brouwer and Fitzpatrick 2000 and 2002, this issue). Note that the upper limit of the reddish yellow mottles in profiles G-5 to G-7 is largely parallel to the soil surface. Light yellow brown and black mottles are found throughout the B-horizon of G-8, indicating a long duration of saturation (Vepraskas *et al.* 1994).

The strong brown mottles that form from the red mottles of the plinthite are well developed in the brown yellow B-horizon on the crest and on the upper part of the B-horizon on the shoulder (G-0 to G-2, 'plinthic on crest'). Along the slope ('plinthic on slope') they are only present in the top 0.15 m of the B-horizon of profile G-5. Soil formation appears to be less rapid in the 'plinthic on slope' than in the 'plinthic on crest' parent material. This may be because the slope induces lateral interflow and surface runoff, at the cost of deeper infiltration and downward progression of soil formation. The reduction of red mottles to strong brown mottles is also less prominent in G-5. It is also possible that the parent materials of the 'plinthic on crest' and 'plinthic on slope' are different (see below).

Red mottles are found more or less everywhere in the B- and C-horizons of all profiles, with the following exceptions. Red (or reddish brown) mottles are not, or no longer, found above 1.0 m depth in profile G-0, where there are (still) some yellowish red mottles. They are also not found above 0.4 m depth in G-1. Red mottles are completely absent from G-8.

Table 1. Selected soil morphology, chemical, and physical properties of typical soil profiles at Gatium

Horizon	Depth (cm)	Matrix colour	Secondary colour	Mottles ^A	Secondary mottles	Cutans ^B	Fe ^C glaeb.	Total C (Leco) (%)	Org. C (%)	pH (1:5 soil:H ₂ O)	EC (dS/m)	Cl (mg/kg)	ESP (%)	Ca	Mg	Na	Exch. cations (mmol _c /kg)	Clay (%)	Silt (%)
<i>G-0, broad crest</i>																			
Ape	0-13	10YR3/2							3.5	4.6	0.49	89	0.6	64	14.4	0.9	4	29	16
Ec	13-20	10YR3/3					fm		1.57	4.8	0.23	40	1.8	55	25.1	2.4	2	35	11
E/Bc	20-30	10YR4/3					fm		1.02	5.4	0.07	10	3.1	48	31.2	3.3	1	38	12
Btc1	30-40	10YR6/8		fcf 10YR5/4		f unsp.	cc		0.92	5.6	0.11	27	3.3	52	41.3	4.1	1	46	9
Btc2	40-60	10YR6/8		cmd 7.5YR5/6		f unsp.	cc		0.52	6.2	0.06	8	4	56	60.6	6.5	2	59	11
Btc3	60-98	10YR7/3		mnp 5YR5/8	5YR5/6				0.14	6.2	0.07	24	6.2	53	70.3	9	1	58	8
Btc4	98-150	10YR7/2		mcp 7.5YR5/8	5YR5/6				0.09	5.8	0.08	39	7.8	39	82.3	13	1	69	7
Btc5	150-195	10YR7/4		mcp 2.5YR4/4	7.5YR5/8				0.1	4.7	0.11	46	9.3	12	56.9	13	1	71	6
Cv1	200-250	10YR7/8		mnp 2.5YR5/6	10YR6/6				0.04	4.4	0.08	59	4.1	1.5	33	5.3	0	47	8
Cv2	270-350	10YR7/4		mnp 10YR8/4	2.5YR4/8				0.02	4.4	0.07	78	7.3	2.3	14.6	2.9	0	33	5
<i>G-3, slightly waning midslope</i>																			
Ap1	0-5	10YR3/2		vfd 7.5YR5/6				2.2		5.4	0.07	40	2.6	19	9.7	2	3	12	26
Ap2	5-15	10YR4/2		vfd 7.5YR5/6															
E	15-35	10YR5/3	10YR4/2	ffd 7.5YR5/6				0.8		5.6	0.05	17	3.2	12	10.9	1.7	1	14	22
Btg1	35-45	10YR5/6		mmd 2.5YR4/6		a 10YR4/3		1.2		6	0.06	21	4.6	29	56	7.6	4	72	12
Btg2	45-65	10YR5/6	10YR6/4&8/2	mmd 2.5YR4/6		a 10YR4/3		0.6		6.3	0.09	44	6.5	28	66.3	9	2	76	11
Btg3	65-85	10YR6/4&8/2	10YR5/6	mmd 2.5YR4/6		a 10YR4/3		0.3		6.3	0.14	114	8.4	22	58.4	9.5	1	72	13
Cg1	85-125	intermediate between Btg3 and Cg2																	
Cg2	125-150	5YR8/1	5YR8/2	cmd 2.5YR4/4	10YR5/4	d 10YR4/2		0.1		6.4	0.2	212	15.2	15	54	11	1	63	21

(continued next page)

Table 1. (continued)

Ap	0-10	10YR4/2	10YR3/2	10YR4/3	vff7.5YR5/6	G-5, <i>midslope</i>	1.9	5.3	0.06	17	2.4	20	10.2	2	2	15	17
E	10-25	10YR5/3	10YR4/3		vff7.5YR5/6	vv	0.7	5.9	0.05	20	3	17	15.6	1.7	1	20	13
Btc1	25-35	10YR5/6			ffd7.5YR5/6	vf	0.7	5.9	0.04	21	3.5	32	44.5	4.4	2	53	14
Btc2	35-40	10YR5/6			ffd7.5YR5/6	cf	0.6	6.1	0.06	32	2.8	31	46	4.1	3	60	9
Btc3	40-65	10YR5/6	10YR4/4		mmp 10R4/6	f 2.5YR2.5/0	0.3	6.3	0.07	36	3.5	29	45.8	4.8	2	55	10
Btcv4	65-115	5YR7/1			mcp 2.5YR4/6	7.5YR6/6	0.2	6.2	0.07	22	4.4	31	58.4	7.2	2	68	7
Btcv5	115-150	5YR7/1			mcp 2.5YR4/6	7.5YR6/6	0.2	5.6	0.07	20	4.6	24	50.6	5.8	1	61	12
<i>G-8, flat valley</i>																	
Apn	0-10	10YR4/2	10YR3/2		vff7.5YR5/6		3.7	5.6	0.1	41	5.2	17	18.2	6.2	3	11	30
En1	10-25	10YR6/2			ffd7.5YR5/6		0.4	6.3	0.09	70	16.3	8.2	6.8	3.1	1	7	30
En2	25-40	10YR6/2			ffd7.5YR5/6	10YR6/4	0.3	6.7	0.13	139	25.3	7.1	8.4	4.8	1	11	30
Bigne1	40-55	10YR6/3			ccd7.5YR5/6	10YR6/6	ff	6.7	0.33	368	18.5	21	33.4	16	2	37	22
Bigne2	55-75	10YR5/3			mcd 7.5YR5/6	10YR6/4	ff	6.4	0.57	786	17.6	39	68.9	31	3	73	9
					ffd7.5YR2/0	c 10YR4/2	0.2										
Bign3	75-100	10YR5/3			mcd 7.5YR5/6	10YR6/4	0.2	6.3	0.67	837	16.2	40	74.8	31	3	75	9
					ffd7.5YR2/0	c 10YR4/2											

^aFirst letter indicates abundance (v = very few <2%, f = few 2-10%, c = common 10-20%, m = many 20-50%); second letter indicates size (f = fine <5 mm, m = medium 5-15 mm, c = coarse 15-30 mm); third letter indicates contrast (f = faint, d = distinct, p = prominent); where abundance, size, or prominence varies within a horizon (cf. Brouwer and Fitzpatrick 2000), only the predominant classification has been mentioned here.

^bFirst letter indicates abundance (f = few <10%, c = common 10-20%, a = abundant 50-90%, d = dominant >90%); unspec. = unspecified

^cFirst letter indicates abundance (v = very few <2%, f = few 2-10%, c = common 10-20%, m = many 20-50%); second letter indicates size (v = very fine <2 mm, f = fine 2-6 mm, m = medium 6-20 mm, c = coarse 20-60 mm, s = stones >60 mm).

Table 2. List of soil features and their definition

Number or numbers in front of a feature indicate the soil system or systems in which the feature is found. The soil systems are defined in the *Results* section of the main text

<i>A- and E-horizon macromorphology</i>	
(all horizons fine sandy loam, massive or (weak) subangular blocky)	
Matrix	
1	Dark brown (10YR3/2&3/3)
1,2,3	Dark grey brown (10YR4/2&4/2-3/2)
1	Brown (10YR4/3)
1,2	Grey brown (10YR5/3-4/2&5/2-4/2)
1,2,3	Pale brownish grey (10YR6/2&6/2-5/2)
Glaebules	
4a,b	Iron glaebules >2% (magnetic and non-magnetic)
<i>B- and C-horizon macromorphology</i>	
Matrix	
4a,b	Grey & light brown (10YR7/2, 7/3, 7/4, 6/4-7/3; 5YR7/1 in lower part of G-5)
1	Brown yellow (10YR6/8, 5/5-4/3, 5/6, 5/6-4/4, 6/6-5/6)
5	White (5YR8/1&8/2)
5/1	White and brown yellow (mixed) (white as above; brown yellow as below)
1	Brown yellow (10YR5/6-6/4)
1,2,3	Brown (10YR5/3) (paler than brown of E-horizon)
1,2,3	Pale brown (10YR6/3)
Mottles	
4/5	Red (with yellow brown margins) (2.5YR5/6, 4/8, 4/4, 4/6)
1,2	Red yellow (5YR5/6, 7.5YR6/6)
1	Strong brown (7.5YR5/6, 5/8, 4/5; 10-7.5YR5/6)
1,2,3	Light yellow brown (10YR 6/4, 6/6)
2,3	Black (7.5YR2/0)
Cutans	
4	Black (2.5YR5/0, 5YR2.5/0)
1	Reddish brown (5YR4/4)
2,3	Pale (light grey, approx. 7.5YR7/1)
5,1	Dark brown (10YR4/3-3/2, 4/2, 5/2-3/2)
Glaebules	
4	Iron glaebules >2% (mostly non-magnetic)
Structure:	
4	Platy
5	Prismatic
<i>All horizons, chemical analyses</i>	
Salt:	
4a	ESP 5–10
5,3	ESP >15 and chloride >200 mg/kg

This and other differences (Table 1) point to the latter profile not having undergone the humid tropical weathering (lateritisation) that the other profiles underwent.

More than 2% ironstone glaebules or gravels are found in the top of the plinthic subsoil on the crest (yellowish brown B-horizon of G-0 to G-2), and virtually throughout the plinthic subsoil on the slope (G-5 to G-7) (Fig. 3c). These glaebules appear to have been formed *in situ*, as they have a relatively angular shape. In the pallid subsoil no glaebules are found, which is to be expected as there is no iron present for them to be formed from. In the alluvial/colluvial profile at G-8, the glaebules in the subsoil appear to have come in from upslope; they are more rounded, suggesting transportation, and glaebules are absent from lower parts of the B-horizon of G-8.

Cutans are prominent in many of the profiles, indicating illuviation from overlying horizons in the past and present (Vepraskas *et al.* 1994). The cutans are particularly abundant in the pallid zone material, where they are dark brown and cover all the ped faces. They are somewhat more pronounced every 0.3–0.6 m laterally, pointing to larger units within the prismatic structure of the pallid zone (cf. Lewis 1985). Brown cutans also occur in profile G-8. Very dark (black) cutans are found throughout the B-horizon of profile G-5 and in the upper parts of the B-horizon of profiles G-6 and G-7 ('plinthic on slope'). In the 'plinthic on crest' material the cutans are less clear but still quite noticeable and both black (2.5YR) and reddish brown (5YR) in colour. The reddish brown cutans are probably conversions of black cutans. This conversion could be related to the general yellowing of the soil from above, showing the same shift in hue from reddish to yellowish as do the converted mottles. As with the mottles, the changes in colour of the cutans indicate that the cutans were not formed under present hydrological conditions, only changed by them. Note that in profiles G-5, 6, and 7 no reddish brown cutans were observed.

The pale (light grey) cutans in the lower parts of profiles G-6 and G-7 appear to be conversions of the black cutans found higher up in the same profiles. The pale cutans, too, show a slight yellowing (from 5YR to 7.5 YR), in addition to a pronounced increase in value relative to the black cutans. This zone of colour change in the cutans borders on the area of very wet conditions in profile G-8, and has an upper boundary which is not parallel to the soil surface (Fig. 3c). The pale colour of cutans in lower parts of profile G-6 and throughout the B-horizon of profile G-7 is most likely caused by present, seasonally saturated, and possibly saline, conditions. These conditions are related to water coming in laterally and/or up from below into profiles G-6 and G-7.

Soil structure of the B- and C-horizons is shown in Fig. 3a. Wherever structure is not indicated as being platey or prismatic, it is angular blocky. In the 'plinthic' materials, the original platey structure is still present in the lower parts of the B- and in the C-horizons. The conversion from platey to angular blocky structure has not progressed as deeply as the conversion from a grey soil matrix with red mottles to a yellowish brown matrix with strong brown mottles (cf. profiles G-0 to G-2, and G-5 to G-7, in Fig. 3a). Within the plinthic subsoils, the conversion of the soil structure appears to take more time than the conversion of the iron minerals. The change in soil structure is probably due to the changes in hydrological circumstances; more extreme wetting and drying causes more shrink-swell, which in turn causes a conversion of the platey structure. The 'onion peel'-type patterns in the mottled zone on the crest indicates that the original platey structure may be traced back to the geologic structure of the pyroclastic parent material (cf. Brouwer and Van de Graaff 1988).

In the pallid zone subsoil layers, white and light grey colours persist where the prismatic structure has already been converted to angular blocky. Arguably, this is because there is

less iron present in the pallid material, causing a change in colour to yellowish brown to take longer.

Growth of grass roots was rather similar in all profiles. In the B-horizons grass roots were mostly along ped faces. Tree root holes of various ages are found in all profiles, at all depths. The field description from the 3.2-m-deep permanent profile pit by Brouwer and Van de Graaff (1988) shows the importance of trees in the development of the yellow duplex soils in this landscape. Where a tree formerly stood the IIB2-horizon extended downwards to 1.3 m rather than only to 1.0 m. And material similar to that in the B2-horizon was found to a depth of 2.3 m along vertical old tree root holes. Remnants of tree roots were also found along major cracks in the mottled parent material at more than 2.5 m depth. Examination of the faces of eight 1.5–3-m-deep inspection pits on the broad crest indicated that former tree root channels in the lower B-horizons can be classified according to their supposed age: in the youngest ones the original root is still present; in older ones the original root is gone but the material filling the root holes is still dark in colour (field texture very fine Sandy Loam to Silt); in the presumably oldest root holes the infill is pale yellow (very fine Sandy Loam to Silty Clay Loam). Many old tree root holes in the pit walls were occupied by lucerne or phalaris roots (J. Brouwer, unpublished data).

Of the soil chemical properties in profiles G-0, 3, 5, and 8 (Table 1), only salt is discussed here as a soil feature. In profile G-0 ('plinthic on crest') exchangeable sodium percentage (ESP) values of 5–10 were found below 0.6 m, i.e. below the brown yellow B-horizon and largely below the first restricting layer at 0.8–1.0 m depth. These elevated ESP values are most likely caused by Na transported in rain. Chloride concentrations were somewhat elevated (59 and 78 mg/kg) only below 2.0 m; higher up in the profile any high chloride concentrations had already been leached. The elevated concentration of chloride at the surface of profile G-0 (89 mg/kg) may be related to application of fertiliser (KCl); the $P(HCO_3^-)$ phosphorus concentration there was also very high (63 mg/kg). In profile G-5 ('plinthic on slope'), both ESP and chloride concentrations were low throughout the profile. This may point to a different origin of the two plinthic parent materials, rather than to higher leaching on the slope (see further discussion below).

In profile G-3 the ESP was 15.2 in the white layers (where the chloride concentration is 212 mg/kg), and 8.4 and 6.5 in the layers immediately above (chloride 114 and 44 mg/kg). This is indicative of storage of cyclic salt in the pallid zone (as found elsewhere in the region; Lewis 1985), and with the gradual removal of the salt where that layer is close to the surface.

Profile G-8 had an ESP >15 (up to 25.3) to within 0.15 m of the surface, caused by saline groundwater moving in from upslope or below. The chloride content was around 800 mg/kg below 0.6 m. It is likely that the lower parts of profiles G-6 and G-7, where no samples were taken for chemical analysis, also had elevated ESPs and chloride concentrations. These profiles show macromorphological signs of saturation in their lower regions, e.g. the pale cutans mentioned above. However, it is not clear whether the paler colour of these cutans is due to saturation by fresh water interflow or by saline groundwater.

For soil classification purposes it is important to note that almost all horizon boundaries other than E-B transitions are gradual. E-B transitions are clear, except over the pallid domain (G-3 and G-4, abrupt) and on the crest and on the flat (G-0 and G-8, gradual).

Soil classification

Only the data necessary for classification of the four typical profiles, G-0, G-3, G-5, and G-8, are presented in Table 1. Their full descriptions along with partial descriptions of the other profiles can be found in Brouwer and Fitzpatrick (2001). The results of the soil

Table 3. Classification of soil profiles down the toposequence at Gatum

Where two classifications are mentioned under Isbell (1996) and Soil Taxonomy (Soil Survey Staff 1996), the second (in brackets) represents a proposed new subgroup for that classification system

prof. #	Factual Key (Northcote 1979)	Australian Soil Classification (Isbell 1996)	Soil Taxonomy (Soil Survey Staff 1996)
G-0	Gn4.34 grading to Dy3.21	Reticulate (Sodic-Reticulate) Eutrophic Yellow Dermosol	Typic (Kandic) Plinthoxeralf
G-1	Dy3.21 grading to Gn4.34	Ferric (Ferric- or Sodic-Reticulate) Eutrophic Brown Chromosol	Typic (Kandic) Plinthoxeralf
G-2	Dy3.21/Db2.21 (Dd2.21)	Reticulate (Sodic-Reticulate) Eutrophic Brown Chromosol	Typic (Kandic) Natrixeralf
G-3	Dy3.21 (Db2.21/Dd2.21)	Mesotrophic Sodosolic Redoxic Hydrosol	Aquic Natrixeralf
G-4	Dy3.21 (Db2.21/Dd2.21)	Mesotrophic Sodosolic Redoxic Hydrosol	Aquic Natrixeralf
G-5	Dy3.21	Reticulate (Ferric-Reticulate) Mesotrophic Brown Chromosol	Typic (Kandic) Plinthoxeralf
G-6	Dy3.21	Reticulate Mesotrophic Brown Chromosol	Typic (Kandic) Plinthoxeralf
G-7	Dy3.41	Bleached-Ferric (Ferric-Reticulate) Mesotrophic Brown Chromosol	Typic (Aquic) Plinthoxeralf
G-8	Dy3.42 (Dd2.42)	Natric Kandosolic Redoxic Hydrosol grading to Kandosolic Salic Hydrosol (Bleached Natric Redoxic Hydrosol)	Typic Natraqualf

classifications are shown in Table 3. For some of the soils it could be useful to define a new subgroup in the Australian Soil Classification system and/or Soil Taxonomy. Such proposed new subgroups are indicated in brackets in Table 3.

Soil classification using Northcote (1979)

All profiles except G-0 qualify as yellow duplex soils (VC rating of the soil matrix of 4 or 2), although G-2 can also be classified as a brown duplex soil (VC rating of 4–5). The texture change at G-0 is less sudden, possibly due to a lesser influence of hill slope processes, leading to classification as a gradational soil.

Where cutans or mottles are very prominent in the B-horizon, their colour was used for an alternative colour classification. Such alternative classifications are given in parentheses in Table 3, and highlight the marked presence of dark cutans or mottles (alternative classifications as brown or dark duplex soils).

All profiles have a hard setting A-horizon and a pedal, mottled B-horizon (first digit 3 or 2 in Northcote's classification, depending on which subdivision they are in). All profiles have an A2-horizon, some bleached throughout (second digit 4), some not bleached at all (second digit 2). All profiles but one have an acid Soil Reaction Trend (last digit 1); profile 8 has a neutral Soil Reaction Trend (last digit 2), probably related to its (lateral) enrichment with salts. The toposequence can be characterised as consisting of hard, pedal, mottled yellow duplex soils, with bleached A2-horizons occurring part way along and at the bottom of the slope.

Australian Soil Classification System (Isbell 1996)

Using the system of Isbell (1996), the upper parts of the toposequence can be described as consisting of (mostly) Reticulate Brown Chromosols, with local ironstone gravel

concentrations. Hydrosols with salt accumulations occur part-way along the slope and also on the valley bottom.

Soil Taxonomy

Soil classifications according to Soil Survey Staff (1996) are also given in Table 3. The toposequence can be described as consisting of Typic Plinthoxerales (upslope), with Aquic Natrimerals occurring part-way along the slope, and Typic Natraqualfs at the bottom of the slope.

The structural approach

As mentioned, in the structural approach the grouping of soil features is done according to the concordance or discordance of soil features grouped in layers, and according to hydrological systems and parent material. It should be noted that the factors that determine the grouping into 'soil systems' and 'domains' are interchangeable; where the origin of the parent material is the most important, soil domains may be defined on the basis of parent material, and soil systems on the basis of hydrology or other soil forming processes, and vice versa. Because this study is concerned with dryland salinity, the hydrology was deemed to be most important and used to define the domains, while parent material was used to define systems and subsystems.

The following domains and soil systems were recognised at Gatum:

- I. *Hydromorphic domain*. Present hydrology; direction and relative importance of flows indicated by the arrows in Fig. 3d; divided into the following 3 hydromorphic soil systems:
 1. Upper hydromorphic soil system, under the influence of the first perched water table;
 2. Middle hydromorphic soil system, under the influence of the second perched water table; and
 3. Lower hydromorphic soil system, under the influence of the deep, permanent, saline water table.
- Cf. the 3 lateral flow levels in Fig. 3d, and in Fig. 2 of Brouwer and Fitzpatrick (2002, this issue).
- II. *Lateritic domain*. Ancient hydrology; flow according to the dotted arrows of low importance in Fig. 3d was much more important at that time than it is now:
 4. Red soil system, showing fossil iron enrichment:
 - (a) 'Plinthic on crest' subsystem (the red or strong brown mottled areas between G-0 and G-2),
 - (b) 'Plinthic on slope' subsystem (the red or red-yellow mottled areas between G-5 and G-7); and
 5. Pallid soil system, showing fossil iron depletion (the area with white or white and brown yellow matrix between G-3 and G-4).

How the other mapped soil features fit into this classification can be seen in Table 2.

Discussion

Soil hydrology

To gain a better understanding of landscape-wide hydrological processes, the interpretation of the soil features described above can be combined with the general information that the topsoils are often waterlogged in winter throughout the landscape, and that the valleys

suffer from severe salinisation. Before this toposequential investigation commenced, we already had reasonably specific knowledge of the soil hydrology on the crest (Brouwer and Van de Graaff 1988). Even so, the following conclusions regarding soil hydrology can arguably be drawn even if one only possesses general hydrological information in addition to the detailed soil macromorphological data. Where additional piezometric data are required to shed more light on the precise hydrological situation this is indicated.

First, fresh, perched water table

As the topsoil is seasonally waterlogged throughout the undulating landscape, downward flow of infiltrating rainwater is most likely restricted close to the soil surface. From initial inspection of the macromorphological features it is not clear at what depth exactly this less permeable layer is to be found. However, one would expect that, on the crest, the major restriction to downward water flow is either at the top of the B-horizon, or at the level where the platey structure still persists (see Carlan *et al.* 1985; Heuvelman and McInnes 1997). The piezometer data clearly indicate that the first restricting layer on the crest is not encountered until approximately 0.8–0.9 m depth (Fig. 4; Brouwer and Van de Graaff 1988), where the platey peds are quite coarse (20–50 mm). A platey structure is still present well above that depth but is, or has become, relatively fine, viz. 5–20 mm. Thus, in the subsoil of the ‘plinthic on crest’ subsystem, the position of the first level on which the infiltrating water perches appears to be where a coarse platey structure is still present, and just below the depth to which the yellowing of the soil matrix has progressed. It is, by and large, equivalent to the depth to which partial changes of non-matrix colours have progressed; cutans have partially changed from black to reddish brown, and mottles have changed from the original red to more yellow or brown (Munsell colour changes from 2.5YR to 5YR or 7.5YR, and even to 10YR) (Fig. 3*a*, *c*, and *d*; Table 2). Positions of restricting layers well into the B-horizon in duplex soils have also been reported by McFarlane and Cox (1992).

Downward movement through this first restricting layer may be through root holes as well as interpedal cracks (Brouwer and Van de Graaff 1988). This has also been mentioned by e.g. McFarlane and Cox (1992) and Smettem *et al.* (1991) for duplex soils occurring in Western Australia and South Australia, respectively, and by Van Noordwijk *et al.* (1991) for acid soils in the humid tropics. There will also be considerable lateral movement or interflow of water over the top of this restricting layer. Lateral movement will be particularly strong where the top of the restricting layer has a significant slope, such as between profiles G-1 and G-2, and where overlying layers are quite permeable. (Note again that the difference in scales used to depict soil depth and surface elevation can lead to layers appearing to slope inwards, while in effect they simply follow the general slope of the surface; compare the mapping of the first restricting layer in Fig. 3*d* here (exaggerated scale of soil depth) with the same line in Fig. 3 of Brouwer and Fitzpatrick (2002, this issue) (soil depth and landscape vertical scales the same).

In the ‘plinthic on slope’ subsoil the precise location of the first restricting layer is more difficult to pinpoint with the available morphological and piezometric data. The parent material of this subsoil appears to be of different origin to that of the subsoil of the ‘plinthic on crest’ subsystem (see below). The dark cutans in the top of the B-horizon show little if any sign of discolouration, and there are no strong brown mottles in the upper part of the B-horizon. The platey structure is still relatively coarse (up to 50 mm) quite close to the surface. In addition the slope of the relatively permeable A- and E-horizons (approximately 0.15 m/day; Brouwer and Van de Graaff 1988), and of the soil surface itself, would favour

interflow and runoff at the expense of deeper infiltration. All this suggests that the restriction to downward flow of water is at approximately 0.40 m depth in the subsoil of the 'plinthic on slope' subsystem. The only argument against this is that the B-horizon has yellowed down to 0.65–0.85 cm depth. Perhaps the different origin and nature of the parent material (more hard iron concretions present) makes the 'plinthic on slope' subsoil yellow more easily than the 'plinthic on crest' subsoil, even though it is not more permeable.

Second, fresh, perched water table

A relatively pale colour of the E-horizon in profiles G-2, 3, and 4, where the old pallid zone is near the surface (Fig. 3a), indicates that the pallid zone also restricts downward movement of water in the soil. This conclusion is supported by illuviation features such as tonguing of E-horizon material into the B-horizon. In addition there is the relatively unaltered state of the pallid zone material quite close to the surface, including original cutan colours (Fig. 3a–c). More importantly, at the upper boundary of the pallid zone there is a very abrupt increase in texture, from fine loamy sand or sandy loam in the E-horizon to heavy clay in the B-horizon. This textural change causes the perching of infiltrating water on top of the pallid zone.

The piezometric data from elsewhere in the landscape also indicate that there is a second perched water table on top of the pallid zone; the piezometer at 3.2 m depth, on top of the pallid zone, had a short period of saturation even during the relatively dry winter of 1987 (Fig. 4; see also Brouwer and Van de Graaff 1988, and Brouwer and Fitzpatrick 2000). Because there is no appreciable salt store above the pallid material, the water table perched on the pallid material is fresh (unpublished data).

Very prominent dark cutans are evidence that downward movement of water through the pallid zone material is mostly along cracks between the prismatic peds. Larger cracks, with even more prominent cutans, occur every 0.3 to 0.6 m. Water movement through the pallid zone may also be partially via old root holes (cf. Johnston *et al.* 1983).

Where the pallid zone comes close to the surface and the first and second perched water tables merge, waterlogging problems are compounded. This occurs at profiles G-2 to G-4, which are located in or near a hillside seep as discussed by Nulsen (1985), although the seep is of slightly different origin from the ones Nulsen describes. It can be assumed that the two perched water tables also merge in the lower parts of profiles G-6 and G-7 (bleaching of originally black cutans), and in all of profile G-8 (a very wet soil).

Third, saline, permanent water table

The presence of a saline groundwater table is evident from the salinity problems in the valleys of the Dundas Tableland. Data from the piezometers indicate that this permanent groundwater table is mainly found in the weathered ignimbrite and pallid material overlying the unweathered ignimbrite. Under the broad crest the ignimbrite can be found at more than 20 m depth, but the saline groundwater table can rise to within 4 m from the surface (Figs 2 and 4). This saline water table could therefore cause saline hillside seeps to occur quite high in the landscape, as high as just below the broad crests.

The high salt contents in profile G-8 indicate that the saline groundwater comes very close to the surface at the foot of the slope. The combined contributions at G-8 of the three water tables that we have identified make this part of the landscape so wet. On present data it cannot be determined whether the water causing the bleaching of the cutans in the lower part of profiles G-6 and G-7 is fresh (interflow over the top of the pallid material) or saline

(rising groundwater). The ESP classes drawn at profiles G-6 and G-7 in Fig. 3d are therefore provisional.

Clearly, knowledge of the precise location of these 3 water tables from piezometric data helps in the interpretation of the complicated macromorphological data. The piezometric data can be used to further detail how water presently flows between different parts of the toposequence, as first inferred from the macromorphological data. However, now that the conclusions of this paper are available regarding the relationships between various macromorphological features and the occurrence of different water tables on the eastern Dundas Tableland, any future hydrological studies in the area do not need to spend quite so much effort on collecting piezometric data.

Soil classification

Soil classification according to the Australian Soil Classification system of Isbell (1996) preserves a lot more information about the toposequence than does classification according to Northcote (1979) (Table 3). Isbell's Chromosols and Dermosols are here comparable to Northcote's duplex soils and gradational soils, respectively. In Isbell's system wetter duplex soils are classified as Hydrosols. When using Isbell, salinisation (marked presence of sodium), the presence of plinthite (indicating a potential perched water table), and the marked presence of ironstone gravel are reflected in the soil names. Information about base status is also included in some of the classifications. Using Isbell's system, some of the profiles, with value rating 5, are classified as Brown, as opposed to Northcote's (1979) Yellow.

However, certain important combinations of features are not catered for in the presently defined subgroups for Chromosols and Dermosols. These include 'Sodic-Reticulate' (profile G-0) and 'Ferric-Reticulate' (profile G-5). In this study, the 'Sodic' indicates the presence of salts that may affect the quality of the watertable. It is therefore proposed that the classification system of Isbell (1996) be expanded to include 'Sodic-Reticulate' and 'Ferric-Reticulate' subgroups for Chromosols and Dermosols. Similarly, it is proposed that a 'Bleached, Natric' subgroup and great group for Redoxic Hydrosols be created (profile G-8) (cf. the full profile descriptions in Brouwer and Fitzpatrick 2002).

Soil Taxonomy (Soil Survey Staff 1996) highlights well some of the main features of the eight Xeralfs. Some have plinthite at depth, causing water to perch at depth (i.e. Typic Plinthoxeralfs). Others have a natric horizon at depth and either waterlogging on top of the B-horizon (Aquic Natrixeralf), or no waterlogging (Typic Natrixeralf). The Typic Natraqualf at the bottom of the toposequence has, as its name indicates, severe waterlogging problems and strongly natric subsoil. However, certain other features such as ironstone gravel concentrations, and certain combinations of features, are not evident from soil names at the subgroup level. In addition, Soil Taxonomy puts rather much emphasis on the dryness of these soils during part of the year (xeric moisture regime at the suborder level). In Australian environments such seasonal dryness of soils is more the rule than the exception, and need not be used as a classification criterion at a high level.

For Soil Taxonomy a new subgroup, Kandic, is proposed, for both Plinthoxeralfs (profiles 0, 1, 5, and 6) and Natrixeralfs (profile 2) (Table 3). Neither group has any other subgroup than Typic at present. For profile 7 the new subgroup of Aquic Plinthoxeralf would appear to be a possibility, but the soil chemical data required to support this are currently unavailable. Due to the very pale colour of the parent material there were problems recognising the presence of redox depletions in profiles 3 and 4. However, it is

reasonable to assume that these Natriferals, which can be saturated for more than 3 months in winter, are Aquic.

Overall the various soil classification systems are fairly good to very good at identifying properties in particular horizons that influence water movement and plant growth. At Gatum these include, for instance, sharp increases in clay content with depth, and horizons with periodic saturation, high salt contents, or plinthite. The emphasis is on soil horizons with particular properties within individual soil profiles. There is less emphasis on identifying links between different profiles and different parts of the landscape. In addition, using only standard soil descriptions it can be difficult or impossible to separate the effects of past and present soil-forming processes, including past and present soil hydrology. For management purposes it can of course be very important to be able to distinguish between the two.

The structural approach

In the structural approach it is not the soil horizons but the soil features that are important. These features were grouped into soil systems and domains that cover various parts of the toposequence. The systems and domains highlight the processes that link different parts of the landscape. Past and present hydrological and chemical processes were successfully separated into the hydromorphic and lateritic domains.

Within the lateritic domain, the 'plinthic on crest' and 'plinthic on slope' subsystems differ in their mottling, cutans and glaebules, in their structure development, and probably in their hydrologic behaviour. The 'onion peel' structure and large cracks, so evident in G-0 and elsewhere on the crest and shoulder (cf. Brouwer and Van de Graaff 1988), are absent in G-5, 6 and 7. Furthermore, profiles G-0 to G-2 are all eutrophic, and profiles G-5 to G-7 all mesotrophic. Arguably this is due to differences in genesis (different ignimbrite flows as parent material) and/or to the presence of a fault between e.g. G-4 and G-5 (cf. Simpson and Woodfull 1994). The presence of a fault between profiles G-4 and G-5 would also explain the outcropping of the pallid zone at G-3 and G-4. The faulting could have occurred during the uplifting and doming of the Tableland. Alternatively, the ignimbrite flow that eventually gave rise to the profiles G-5, 6, and 7 may have progressed along an ancient valley, in the side slopes of which the pallid material of profiles G-3 and 4 was already cropping out.

As mentioned, the various soil systems and subsystems delineate areas under the influence of the same landscape and soil-forming processes. Thus soil layers or horizons that are part of, or close to, the saline, lower hydromorphic soil system will presently experience, or develop, salinity problems. The layers or horizons that overlie the pallid soil system are likely to have severe waterlogging problems, even if at the time of the field observations that is not evident. Being able to indicate with some precision the extent of the influence of a particular soil system, and its defining processes, makes it possible to formulate and refine site-specific land management recommendations. If similar soil domains and soil systems are found to occur in other areas, similar land use problems may arise, and similar solutions to those problems may be applicable. That is a good starting point for further discussions and possible actions in such new areas.

Note, however, that the different soil systems and domains can expand or contract, as the processes that define them occur over greater or smaller volumes of soil; this must be taken into account when evaluating the effects of possible changes in land management. The soil systems and domains also show some overlap. For instance, remnants of soil system 4, i.e. pieces of ironstone gravel from the lateritic domain, are found in upper parts of the

solum that are now part of soil systems 1, 2, and 3 (as defined by the presence of the different water tables). At the same time, under the influence of these water tables, the soil layers and horizons of the plinthic and pallid soil systems are becoming browner in colour. Drawing precise boundaries between soil systems can therefore be arbitrary. It is better to indicate the overlap, and remark that, in the area of overlap, all the processes used to define the different soil systems may play a role.

Conclusions and recommendations

The macromorphological features of the toposequence were effectively described during a single field visit. Analysis of the data using both soil classification and the structural approach gave complementary information on *in situ* limitations to land use and on landscape-wide processes. When the macromorphological data were combined with farmers' knowledge and limited piezometric data, and analysed using the structural approach, the following conclusions were drawn:

1. A refinement of the soil feature–system–domain grouping method of Fritsch and Fitzpatrick (1994), based on Boulet *et al.* (1982) and Fritsch *et al.* (1992) and also called the structural approach, was used successfully at Gatum to distinguish between the effects of past and present hydrological processes on soil macromorphology.
2. Waterlogging of the surface horizons was not always caused by a restricting layer occurring at the top of the B-horizon. This is contrary to commonly held views. Rather, ponding of infiltrating rainwater most likely occurred on the top of the unaltered part of the old mottled zone where platy structure is still coarse, and above which there had been at least a partial change in non-matrix soil colours: from black to reddish brown in the cutans, and from red to reddish yellow or strong brown in the mottles. On one type of plinthite soil at Gatum, 'plinthic on crest', the colour change had progressed well into the B-horizon, to between 0.6 and 1.0 m depth (G-0, 1, and 2). On another type of plinthite soil on different parent material, 'plinthic on slope', the first major restriction to downward flow of water was found close to the top of the B-horizon, at approximately 40 cm depth (G-5, 6, and 7).
3. Tree root holes and interpedal cracks were significant channels for water flow through the first restricting layer. Tree root holes were found in all profile pits and in all parts of the landscape.
4. A second perched water table, also non-saline, occurred on top of the pallid material. On the broad crests this was at a depth of approximately 3 m. On slopes where the pallid material came to the surface, the first and second perched water tables merged and caused the formation of a hillside seep.
5. Hillside seeps occurred even where the surface topography gave no indication of their likely presence.
6. The permanent, saline water table came close to the surface in the lower parts of the landscape, causing salting problems. These problems were exacerbated in those areas where there were also waterlogging problems caused by water originating from the two perched water tables. The permanent water table was on top of the bedrock. On the broad crests bedrock was at a depth of >20 m, but the permanent water table there can seasonally rise to within 4 m of the surface. This explains why the permanent water table caused saline seeps quite high in the landscape.

These conclusions regarding the hydrological processes have mostly been drawn indirectly, i.e. based on the macromorphology of the soil profiles. Quantification of the

relationships between macromorphology and hydrology requires more direct observations of water movement, e.g. by piezometry. The added value of such data, in the study here presented, is discussed in Brouwer and Fitzpatrick (2002, this issue). However, the present paper already provides results that can be used in the extrapolation of soil hydrological processes from one part of the landscape to another. It also demonstrates what is required for this extrapolation, i.e. detailed soil macromorphological data and their interpretation according to the structural approach.

The soil macromorphological data, farmers' knowledge, and limited piezometric data together allow formulation of the following concrete land management recommendations:

1. To alleviate waterlogging and salinity problems, water that has infiltrated into the soil may be intercepted before it enters areas prone to waterlogging or before it picks up salt. One option for doing this is to increase transpiration by vegetation. Perennial pastures have been suggested for this purpose, but care has to be taken that the pastures do not reduce runoff more than they increase evaporation: where they do decrease runoff too much, the net result will be increased recharge in spite of increased evaporation (Brouwer 1989). If trees are used to increase transpiration, they may be used to greatest effect where the soil is wettest, i.e. above hillside or footslope seeps.
2. Artificial drains are another option for intercepting excess water in the soil. The macromorphological and hydrological data presented here indicate that such drains can be located as deep as 1.0 m, rather than only at 0.3 m, at least in some soils on the Dundas Tableland. Drains intercepting interflow will be located most effectively immediately upslope of hillside and footslope seeps. Either subsurface or surface drains can be used, depending on the circumstances.
3. Deep infiltration to the groundwater probably takes place via old tree root holes as well as via interpedal cracks. Areas where trees used to stand may therefore be areas of relatively high recharge to the groundwater. Such areas could therefore also be targeted for artificial drainage or tree planting.
4. On the flat broad crests, areas of high recharge may be those that have relatively permeable horizons overlying a restricting layer. The more permeable a horizon, the more rapid the lateral flow to root holes and cracks that penetrate the restricting layer.

Acknowledgments

Robert van de Graaff initiated the Gatum project, on the property of Neil and Sue Lawrance. Ian Dreher, Bruce Trebilcock, Phil Miles, and Bill Coe put in long hours at the field site. Advice on various aspects was received from several officers of the Department of Conservation, Forests and Lands (Portland Region) and the Dept of Agriculture and Rural Affairs (South-west Region). Funding for the project was provided by the National Soil Conservation Program, the Australian Wool Corporation and the Salinity Program of the Government of Victoria. CSIRO provided a three-month research fellowship to the first author. Phil Davies and Philippa Butterworth assisted during the preparation of the manuscript, while Greg Rinder drafted the figures. The librarians at the Department of Conservation, Forests and Lands and at CSIRO in Adelaide were extremely helpful. Ray Isbell, Robert van de Graaff, Phil Dyson, Mal Lorimer, Harm van Rees, Jon Fawcett, and two anonymous referees commented on drafts of this paper. To all these, our sincere thanks.

References

- Boulet R, Humbel FX, Lucas Y (1982) Analyse structural et cartographie en pédologie: II Une méthode d'analyse prenant en compte l'organisation tridimensionnelle des couvertures pédologiques. *Cahiers ORSTOM, série Pédologie* **19**, 323–339.
- Brouwer J (1989) Reducing groundwater recharge—is increasing plant water use enough? *Salt Force News* (Salinity Newsletter of the Government of Victoria) **12**, 8–9.
- Brouwer J, Fitzpatrick RW (1998) Relations between soil macro-morphology and current soil hydrology in a toposequence in SE Australia. In 'Proceedings of the International Soil Science Society Congress'. Montpellier, France. Symposium No. 15. CD-ROM.
- Brouwer J, Fitzpatrick RW (2000) Characterisation of nine soils down a salt-affected toposequence near Gatum on the Dundas Tablelands in south-west Victoria: application of the structural approach for constructing soil-water-landscape models. CSIRO Land and Water Technical Report no. 21/00. <http://www.clw.csiro.au/publications/technical2000/tr21-00.pdf>
- Brouwer J, Fitzpatrick RW (2002) Restricting layers, flow paths and correlation between duration of soil saturation and soil morphological features along a hillslope with an altered soil water regime in western Victoria. *Australian Journal of Soil Research* **40**, 927–946
- Brouwer J, van de Graaff RHM (1988) Readjusting the water balance to combat dryland salting in southern Australia: changing the hydrology of a texture contrast soil by deep ripping. *Agricultural Water Management* **14**, 287–298.
- Carlan WL, Perkins HF, Leonard RA (1985) Movement of water in a plinthic paleudult using a bromide tracer. *Soil Science* **139**, 62–66.
- Cox JW, Fritsch E, Fitzpatrick RW (1996) Interpretation of soil features produced by ancient and modern processes in degraded landscapes: VII. Water duration. *Australian Journal of Soil Research* **34**, 803–824.
- Cayley RA, Taylor DH (1997) Grampians special map area geological report. Geological Survey of Victoria Report 107, 150 pp.
- Dahlhaus PG, MacEwan RJ, Nathan EL, Morand VJ (2000) Salinity on the southeastern Dundas Tableland, Victoria. *Australian Journal of Earth Sciences* **47**, 3–11.
- Fitzpatrick RW, Fritsch E, Self PG (1996) Interpretation of soil features produced by ancient and modern processes in degraded landscapes: V. Development of saline sulfidic features in non-tidal seepage areas. *Geoderma* **69**, 1–29.
- Fitzpatrick RW, McKenzie NJ, Maschmedt D (1999) Soil morphological indicators and their importance to soil fertility. In 'Soil analysis: an interpretation manual'. (Eds K Peverell, LA Sparrow, DJ Reuter) pp. 55–69. (CSIRO Publishing: Melbourne)
- Fritsch E, Fitzpatrick RW (1994) Interpretation of soil features produced by ancient and modern processes in degraded landscapes. I. A new method for constructing conceptual soil-water-landscape models. *Australian Journal of Soil Research* **32**, 889–907.
- Fritsch E, Peterschmitt E, Herbillon AJ (1992) A structural approach to the regolith: identification of structures, analysis of structural relationships and interpretations. *Sciences Géologiques, Bulletin* **45**, 77–97.
- Gibbons FR, Downes RG (1964) A study of the land in south-western Victoria. Soil Conservation Authority, Kew, Victoria, Australia.
- Heuvelman WJ, McInnes KJ (1997) Spatial variability of water fluxes in soil: a field study. *Soil Science Society of America Journal* **61**, 1037–1041.
- Isbell RF (1996) 'The Australian soil classification.' (CSIRO Publishing: Melbourne)
- Jenkin JJ (1981) Terrain, groundwater and secondary salinity in Victoria, Australia. *Agricultural Water Management* **4**, 143–171.
- Johnston CD, Hurlle DH, Hudson DR, Height MI (1983) Water movement through preferred paths in lateritic profiles of the Darling Plateau, Western Australia. Groundwater Research Technical Paper No. 1, CSIRO, Australia.
- Joyce EB (1991) The west Victorian Uplands of south-eastern Australia: Origin and History. *Earth Science Processes and Landforms* **17**, 407–418.
- Lewis MF (1985) Factors affecting the development of dryland salinity in a catchment on the Dundas Tableland, western Victoria. Land Protection Service, Dept of Conservation, Forests and Lands, Internal Report, Melbourne, Vic.
- McDonald RC, Isbell RF, Speight JG, Walker J, Hopkins MS (1990) 'Australian soil and land survey field handbook.' 2nd edn. pp. 87–183. (Inkata Press: Melbourne)

- McFarlane DJ, Cox JW (1992) Management of excess water in duplex soils. *Australian Journal of Experimental Agriculture* **32**, 857–864.
- Milnes AR, Bourman RP, Northcote KH (1985) Field relationships of ferricretes and weathered zones in southern South Australia: a contribution to 'laterite' studies in Australia. *Australian Journal of Soil Research* **23**, 441–465.
- Nathan E (1999) Dryland salinity on the Dundas Tableland: a historical appraisal. *Australian Geographer* **30**, 295–301.
- Northcote KH (1979) 'A factual key to the recognition of Australian soils.' (Rellim Technical Publications: Glenside, S. Aust.)
- Nulsen RA (1985) Hillside seepages. *Journal of Agriculture—Western Australia* **26**, 128–129.
- Quinn NH (1997) Regolith of the Balmoral 1:100 000 sheet, western Victoria. BSc (Hons) thesis, University of Melbourne (unpubl.).
- Rayment GE, Higginson FR (1992) 'Australian laboratory handbook of soil and water chemical methods.' (Inkata Press: Sydney)
- Rinder GE, Fritsch E, Fitzpatrick RW (1994) Computing procedure for mapping soil features at sub-catchment scale. *Australian Journal of Soil Research* **32**, 909–913.
- Sibley GT (1967) A study of the land in the Grampians area. Soil Conservation Authority of Victoria, Kew, Vic.
- Simpson CJ (1997) Rocklands volcanics group. (Eds RA Cayley and DH Taylor) Grampians Special Map Area Geological Report. Geological Survey of Victoria Report 107, pp. 61–66.
- Simpson CJ (1998) Rocklands volcanics. Geological Survey of Victoria, Unpublished Report 1998/13.
- Simpson C, Woodfull CJ (1994) New field evidence resolving the relationship between the Grampians Group and the Rocklands Rhyolite, western Victoria. *Australian Journal of Earth Sciences* **41**, 621–624.
- Smettem KR, Chittleborough DJ, Richards BG, Leaney FW (1991) The influence of macropores on runoff generation from a hillslope soil with a contrasting textural class. *Journal of Hydrology* **122**, 235–252.
- Soil Survey Staff (1993) 'Soil survey manual.' USDA Handbook No. 18. (US Government Printing Office: Washington, DC)
- Soil Survey Staff (1996) 'Keys to soil taxonomy.' 7th edn. USDA. (US Government Printing Office: Washington, DC)
- Spencer-Jones D (1965) The geology and structure of the Grampians area. Geological Survey of Victoria, Memoir 25, Melbourne, Vic.
- Van Noordwijk M, Widiyanto, Heinen M, Hairiah, Kurniatun (1991) Old tree root channels in acid soil in the humid tropics: Important for crop root penetration, water infiltration and nitrogen management. In 'Plant–soil interactions at low pH'. (Eds RJ Wright, VC Baligar, RP Murrmann) pp. 423–430. (Kluwer: The Netherlands)
- Vepraskas MJ, Wilding LP, Drees LR (1994) Aquic conditions for Soil Taxonomy: concepts, soil morphology and micromorphology. In 'Soil micromorphology: Studies in management and genesis. Proceedings IX International Working Meeting on Soil Micromorphology'. Townsville, Australia, July 1992. *Developments in Soil Science* **22**, 117–131. (Elsevier: Amsterdam)

Manuscript received 22 April 1998, accepted 15 April 2002